

A Conceptual Feasibility Study of Solo Personal Spacecraft for Independent Human Space Travel

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Manuscript type: Analytical review and concept study

Date: 30 June 2026

Language: English en-GB

Scope note: This manuscript evaluates whether a *single-occupant, personally controlled spacecraft* can be engineered and operated with acceptable technical, medical, legal, and economic rigour. It is framed as a peer-review-style concept paper, not as a certification or safety approval document. All quantitative values are stated in SI units unless an official source is quoted in dual units. [1]

Executive Summary

Independent solo travel in space is **not** analogous to solo travel in aviation or marine leisure craft. In physical terms, “free travel” in space is constrained by launch energy, delta-v, life-support mass, radiation environment, orbital traffic, communication latency, deorbit obligations, and rescue limits. In legal terms, private activity in space is not free from state control: under the Outer Space Treaty, non-governmental space activities require national authorisation and continuing supervision, and launching states retain international responsibility and liability for space objects. [2]

The strongest near-term conclusion is that a **one-person spacecraft is technically conceivable only in a tightly bounded mission class**, most plausibly as a derivative capsule-and-service-module vehicle launched on already certified or certifiable commercial infrastructure. The most credible mission envelope is a **3–7 day low-Earth-orbit sortie**, with a more demanding but still conceptually possible **10–21 day cislunar free-return mission** if the vehicle has robust communications, docking capability, radiation storm-shelter provisions, and substantial ground support. By contrast, a truly solo interplanetary mission is presently not a responsible design target because communication delays, radiation burden, autonomous medical care gaps, exercise and habitability needs, and the absence of realistic rescue options all become mission-defining hazards. [3]

For short missions, the recommended architecture is **not** a fully closed-loop deep-space ship. It is a compact pressurised crew module with partial regeneration only where the mass and reliability trade justifies it, backed by storable-propellant service propulsion, solar-array power, batteries, passive-plus-active thermal control, star-tracker and inertial navigation, and an IDSS-compatible docking interface. This judgement follows directly from existing precedent: Mercury and Vostok proved solitary occupancy; Orion demonstrates current deep-space human-rating practice; ISS Environmental Control and Life Support System work demonstrates what is possible in regenerative life support; and modern small-spacecraft avionics and sensors show the degree of miniaturisation now available. [4]

The economics are prohibitive for any literal “personal ownership” model. Public programme data show that completing human-spaceflight transportation systems already entails **multi-billion-dollar development and certification effort**: NASA reports Commercial Crew Transportation Capability contract awards totalling US\$6.8 billion, and GAO reported Orion development costs approaching US\$10 billion. Public Falcon 9 pricing, at US\$74 million per launch as listed by SpaceX, is only a lower-bound cargo analogue and does not capture human-rating, mission control, abort, recovery, insurance, or certification overhead. A serious solo-spacecraft programme therefore belongs not to the consumer market but to the domain of sovereign, quasi-sovereign, or extremely high-capital institutional actors. [5]

Accordingly, the publishable conclusion of this paper is that **solo personal space travel is conditionally feasible only as a highly regulated, professionally operated, capital-intensive expeditionary system**, not as a consumer “free travel” capability. The paper therefore recommends a phased roadmap: first, LEO independent sorties; second, cislunar free-return missions with docking contingency; third, only after major advances in radiation protection, autonomous medicine, and Earth-independent operations, consideration of longer deep-space solo missions. [6]

Abstract and Keywords

Abstract

This paper examines the technical and operational feasibility of free solo travel in space using a personal spacecraft. Because no off-the-shelf personal spaceship currently exists, the study synthesises primary and official sources from NASA, FAA, UNOOSA, ESA, GAO, COSPAR, and SpaceX to derive a reference architecture and mission envelope. Historical precedent is drawn from single-occupant spacecraft such as Mercury and Vostok, while current benchmarks are drawn from Orion, ISS regenerative life support, modern small-spacecraft guidance, navigation and control, international docking standards, and contemporary radiation protection and human-systems standards. The analysis finds that a single-occupant spacecraft is conceivable for short-duration LEO

missions and, with substantially greater complexity, for cislunar free-return operations; however, truly autonomous interplanetary solo flight is not presently supportable within acceptable human-safety and programme-risk bounds. The dominant barriers are not only propulsion and mass fraction, but also radiation protection, autonomous medical care, communication latency, rescue impossibility, legal supervision, debris obligations, and cost. A notional cislunar one-person spacecraft is proposed with an on-orbit wet mass of roughly 9–10 t, average electrical demand near 2 kW, storable-chemical primary propulsion, solar-array power, partial-loop life support, no nominal EVA, and mandatory docking capability. The study concludes that solo personal spaceflight is best understood as an expeditionary engineering problem requiring institutional-grade infrastructure rather than as an unconstrained personal mobility technology. [7]

Keywords

Solo spacecraft; personal spaceship; human spaceflight; cislunar transportation; life support; space radiation; autonomous navigation; commercial space regulation; orbital debris; mission architecture. [8]

Introduction and Literature Review

Framing the Research Question

The phrase “*travel in space freely alone using personal spaceship*” is rhetorically attractive but technically imprecise. In astronautics, “free” cannot mean unbounded motion: spacecraft remain governed by orbital mechanics, propulsion limits, life-support logistics, communications networks, and national and international law. This paper therefore reframes the question into a researchable form: **what spacecraft architecture, mission profile, and governance model would make single-occupant human space travel technically credible, and what mission classes remain inadmissible or premature?** [9]

This study adopts five explicit assumptions because the prompt leaves key mission parameters unspecified. First, the vehicle carries **one trained occupant**. Second, three mission classes are assessed: **LEO sortie** 3–7 days, **cislunar free-return** 10–21 days, and **interplanetary evaluation case** 180+ days. Third, the spacecraft is assumed to be launched by an external launch vehicle that provides ascent and either orbital insertion or translunar injection; the paper studies the spacecraft, not the launcher. Fourth, payload demand is limited to **100 kg returnable or mission equipment mass**, mirroring the scale of modern crew capsules rather than cargo ships. Fifth, the design objective is **survival and mission completion with professional-grade safety practice**, not tourism-level simplification. These assumptions are consistent with Orion’s current deep-space benchmark of four crew for 21 days, the structure of Commercial Crew systems, and NASA human-system standards. [10]

Prior Art and Historical Precedents

A meaningful literature review begins by noting that **solo human spacecraft are historically real**, but they have never been personal in the consumer sense. Project Mercury, the first US human-spaceflight programme, flew six crewed missions from 1961 to 1963 with a single astronaut onboard; its aims were to orbit a person, evaluate human function in space, and recover both astronaut and spacecraft safely. Vostok 1 carried Yuri Gagarin for a 108-minute orbital mission, with re-entry controlled by onboard computing. The historic lesson is therefore double-edged: solitary occupancy is possible, but it has always required large national industrial systems, launch infrastructure, and recovery operations. [11]

Modern crewed design precedent is captured less by Mercury than by Orion and Commercial Crew. NASA's Orion benchmark is especially useful because it is a deep-space, not merely LEO, capsule. NASA reports Orion's crew module habitable volume as **9.34 m³**, pressurised volume as **19.56 m³**, service-module electrical power as **11 kW**, usable propellant as **8,618 kg**, and nominal mission duration as **21 days** for four crew. The important design implication is that even with half a century of technology gains, deep-space crew transport still demands a substantial service module, significant propellant, and large safety margins. [12]

Life support literature points in a similarly conservative direction. NASA's ISS ECLSS has demonstrated roughly **98% water recovery**, while NASA's ISS explanatory material reports approximately **90% water recovery** in the baseline station system and oxygen generation via electrolysis. NASA also reports that each crew member needs about **one US gallon of water per day** for consumption, food preparation, and hygiene, and about **1.83 lb of oxygen per day**. These are powerful benchmarks because they show both what regenerative systems can achieve and how unforgiving consumables remain when scaled over time. [13]

The literature also shows that recycling is not always the right answer. Jones's life-cycle-cost analysis of recycling versus resupply found that resupply systems have low mass and cost for short missions, while recycling systems carry high initial mass and development cost but lower time-dependent logistics. In that analysis, the life-cycle-cost breakeven for a Mars-transit-style recycling system appeared around **313 days**, much longer than the short solo missions examined here. Thus, for 3–21 day missions, *partial closure* is usually a more rational choice than a fully regenerative loop. [14]

Guidance, navigation, and control literature shows strong technological readiness for LEO and improving autonomy beyond it. NASA's 2026 small-spacecraft GNC assessment describes mature combinations of IMUs, Sun sensors, GPS/GNSS, and star trackers for Earth-orbiting spacecraft, while also noting that deep-space navigation still depends heavily on radio transponders and the Deep Space Network. It concludes that future deep-space navigation will need increasingly autonomous methods, combining optical navigation, inertial sensing, clocks, and Earth-independent decision-making. This

is central to the solo-spacecraft question because a lone operator cannot rely on continuous real-time ground intervention. [15]

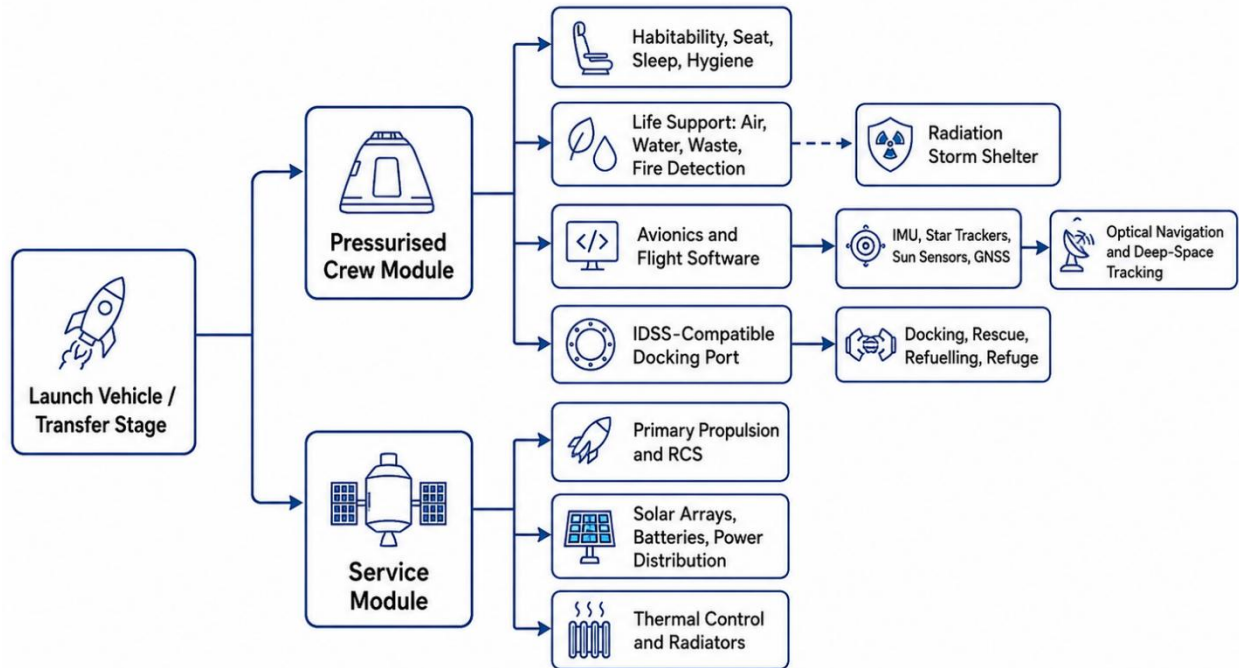
Propulsion literature is similarly mature in taxonomy but not in offering a consumer-like solution. NASA's in-space propulsion reviews separate chemical, electric, and propellant-less systems; NASA's SEP studies examine tens to hundreds of kilowatts for exploration; NASA and DARPA's DRACO work illustrates continuing interest in nuclear thermal propulsion; and ACS3 demonstrates that solar sails remain propulsion demonstrations rather than crew transport workhorses. VASIMR and fusion or beamed concepts remain research topics, not practical near-term foundations for a privately operated one-person spacecraft. [16]

The law-and-policy literature gives the final framing constraint. Under Articles I, VI, VII, VIII, and IX of the Outer Space Treaty, outer space is free for exploration and use by states, but non-governmental activities require authorisation and continuing supervision; launching states remain internationally liable; and activities must avoid harmful contamination and harmful interference. The Rescue Agreement, Liability Convention, and Registration Convention extend this governance architecture. A "personal spaceship" is therefore not legally comparable to a privately owned yacht travelling in ungoverned high seas: it remains embedded in state responsibility, registration, liability, and supervision. [17]

Reference Architecture for a Solo Spacecraft

System Architecture and Design Philosophy

The most defensible reference design is a **compact pressurised capsule plus service module**, rather than a winged vehicle, spaceplane, or "free-flying RV". Capsules remain mass-efficient, aerothermally robust, and operationally simple for launch abort, re-entry, splashdown or land touchdown, and passive safe modes. Orion's architecture illustrates the continuing validity of this arrangement: the crew module handles habitation and re-entry, while the service module supplies propulsion, fluids, power, and thermal control. For a solo spacecraft, the logic is the same even if the mass is reduced. [18]



Conceptual architecture of a one-person spacecraft showing links among the crew module, service module, propulsion, life support, navigation, docking, power, thermal control, and emergency shelter systems.

Figure 1. Reference system relationship diagram for a single-occupant spacecraft. The architecture is derived from Orion-style module separation, ISS-derived life-support practice, NASA GNC state-of-the-art guidance, DSN communications dependence beyond LEO, and IDSS-compatible docking concepts. [19]

A crucial design choice follows from the literature: **nominal EVA should be excluded**. NASA describes EVA suits as “miniature spacecraft”, and PLSS development remains a major subsystem in its own right. Solo EVA removes the possibility of buddy rescue, sharply increases depressurisation and tethering risk, and expands consumables, thermal-control, and training requirements. Therefore, the reference personal spacecraft should be designed for internal maintenance only, with EVA capability held to contingency status or omitted entirely. [20]

Notional Mass, Volume and Materials Budget

The literature does not provide a ready-made single-occupant deep-space capsule, so the following mass budget is an **author-derived synthesis** benchmarked against Orion, ISS life-support quantities, NASA materials literature, and standard crewed-spacecraft

subsystems. It intentionally excludes the launch vehicle and treats any ascent abort tower as a separate pre-orbit mass penalty. Orion's launch abort system alone masses **7,711 kg**, demonstrating how expensive crew safety is in mass terms. [21]

Table 1. Author-derived notional mass budget for a one-person cislunar free-return spacecraft. It is scaled from Orion's mass split and from NASA consumables data for water, oxygen, and packaged food. It should be read as a systems-study estimate, not a certified design. [22]

Subsystem	Notional cislunar solo mass
Crew pressure vessel, primary structure, TPS and landing structure	1,550 kg
Cabin outfitting, seats, restraints, stowage, hygiene, sleep system	350 kg
Avionics, computing, flight software hardware, C&DH	170 kg
GNC sensors and actuators	220 kg
ECLSS hardware	420 kg
Electrical power subsystem	310 kg
Thermal control subsystem	230 kg
Communications, tracking, antennas	130 kg
Docking interface and hatch mechanisms	160 kg
Emergency pressure garments / contingency EVA interfaces	90 kg
MMOD and radiation augmentation, storm-shelter provisions	380 kg
Growth allowance, spares, inspection margin	590 kg
Dry mass subtotal	4,600 kg
Consumables for 21 days, one occupant	130–180 kg
Usable propellant for RCS and mission manoeuvres	4,000–4,500 kg
On-orbit wet mass	8,730–9,280 kg

For habitable volume, a naive linear scaling of Orion would suggest that one occupant might need only around **2.3 m³** of habitable space, because Orion provides **9.34 m³ for four crew**. That would be a mistake. Solo occupancy increases psychological and operational dependence on interior ergonomics, visibility, exercised movement, hygiene, and work/rest segregation. A more defensible target is **6–8 m³ habitable volume** for a

cislunar one-person craft and **3–4 m³** for a short LEO sortie vehicle. This is an engineering inference from Orion volume, NASA human-factors standards, and the known need for exercise and workload management on longer missions. [23]

Materials selection should favour **aluminium-lithium alloys** or hybrid metallic-composite structures for the pressure vessel and major load paths, because NASA materials literature notes Al-Li weight savings of **10% or more** over conventional aerospace aluminium. Secondary and non-pressure structures may exploit carbon-fibre composites where inspection, fracture control, and crashworthiness are acceptable. For micrometeoroid and orbital debris protection, Whipple or stuffed-Whipple-type shielding remains the baseline, while Earth-entry protection may use ablative or tiled systems. Orion, for example, uses a pressure-vessel cover with about **1,300 silica-fibre TPS tiles** on the backshell. [24]

A helpful public-domain benchmark graphic for contemporary capsule architecture is NASA’s **“Orion By the Numbers”** mission graphic, which provides an exploded view of the crew module, service module, adapter, and launch abort system. [25]

Propulsion Trades

For a solo spacecraft, propulsion must be evaluated against four criteria: abortability, restart reliability, mission duration, and operational simplicity. Chemical propulsion remains dominant for crew safety because it can provide prompt thrust for ascent abort, rendezvous braking, and deorbit or return manoeuvres. Electric propulsion is attractive for efficiency but weak for crew contingencies. Nuclear and advanced concepts are highly relevant for the literature review, but not for a near-term “personal” vehicle. [26]

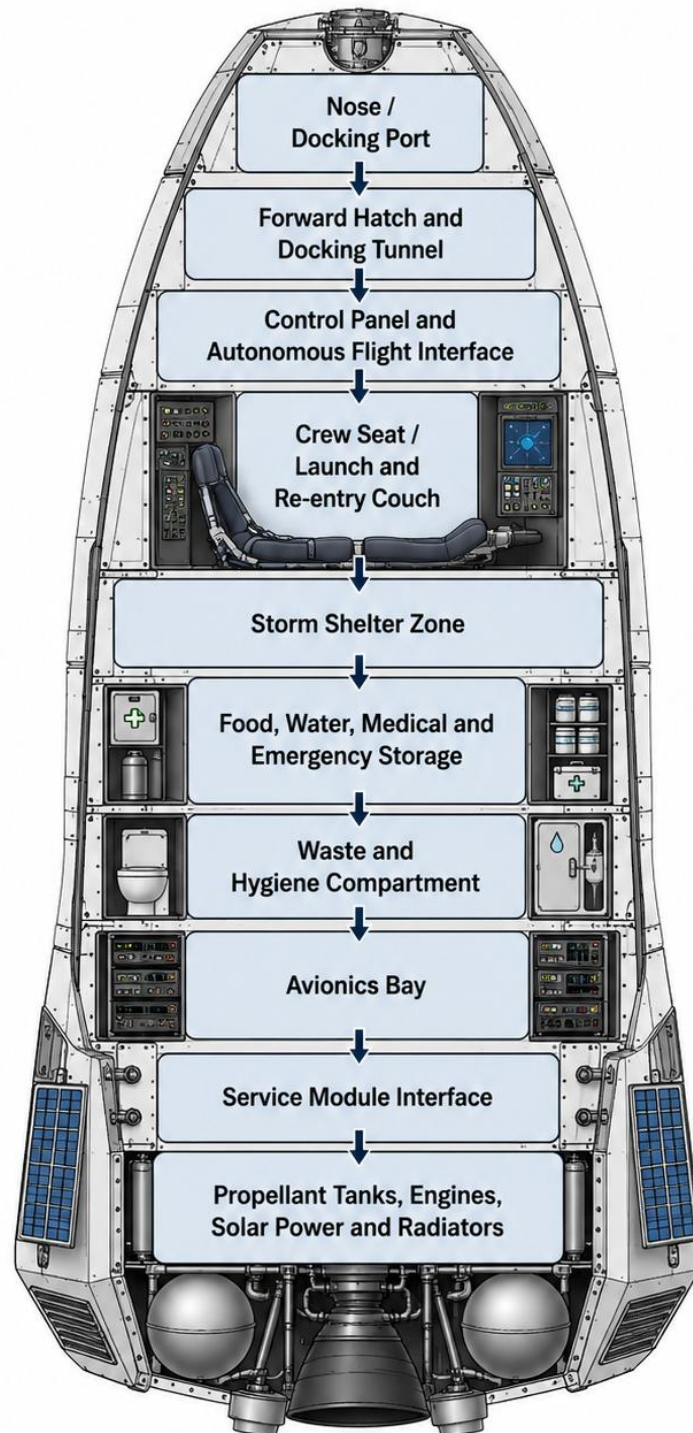
Table 2. Comparative propulsion assessment for a solo personal spacecraft. Values and maturity judgements synthesise NASA’s in-space propulsion assessments, SEP studies, DRACO materials, ACS3, and VASIMR/fusion concept literature. [16]

Propulsion option	Typical specific impulse	Thrust character	Maturity and official status	Suitability for solo spacecraft
Storable chemical monopropellant/bipropellant	~220–340 s	High, immediate, restartable	Mature and widely used in crewed and robotic spacecraft	Baseline choice for RCS, rendezvous, deorbit and cislunar corrections

Propulsion option	Typical specific impulse	Thrust character	Maturity and official status	Suitability for solo spacecraft
Cryogenic chemical	up to ~450 s	High	Mature in upper stages, operationally complex for long coast because of boil-off	Good for launcher stages; less attractive for owner-operated standalone vehicle
Hall or ion electric propulsion	~1,000–3,000+ s	Very low continuous thrust	Mature for robotic spacecraft and cargo missions	Useful only as an auxiliary tug or cargo stage, not as primary crew safety propulsion
Nuclear thermal propulsion	often cited around ~850–900 s target	High	In active development; NASA–DARPA DRACO is a demo, not an operational crew system	Long-term exploration option, not near-term personal use
Solar sail and other propellant-less systems	effectively propellantless	Very low acceleration	Demonstration stage; ACS3 is a technology demonstrator	Unsuitable for prompt human manoeuvre or rescue
VASIMR, fusion, beamed concepts	Concept-dependent, potentially very high	Varies	Research and NIAC-class concepts; not operational	Research only

The recommended baseline is therefore a **storable chemical propulsion service module**. It is less efficient than cryogenic or electric alternatives, but it is simpler to store over multi-day missions, easier to restart repeatedly, and more compatible with crew safety

doctrine. The likely role for electric propulsion in a solo-human architecture is not onboard primary manoeuvre, but rather as a pre-deployed tug, cargo ferry, or orbital transfer stage that is separate from the occupied spacecraft. [27]



Internal layout concept for a compact solo spacecraft, with the astronaut's seat positioned close to the storm shelter and emergency stores for survivability.

Figure 2. Personal Spacecraft Module Layout

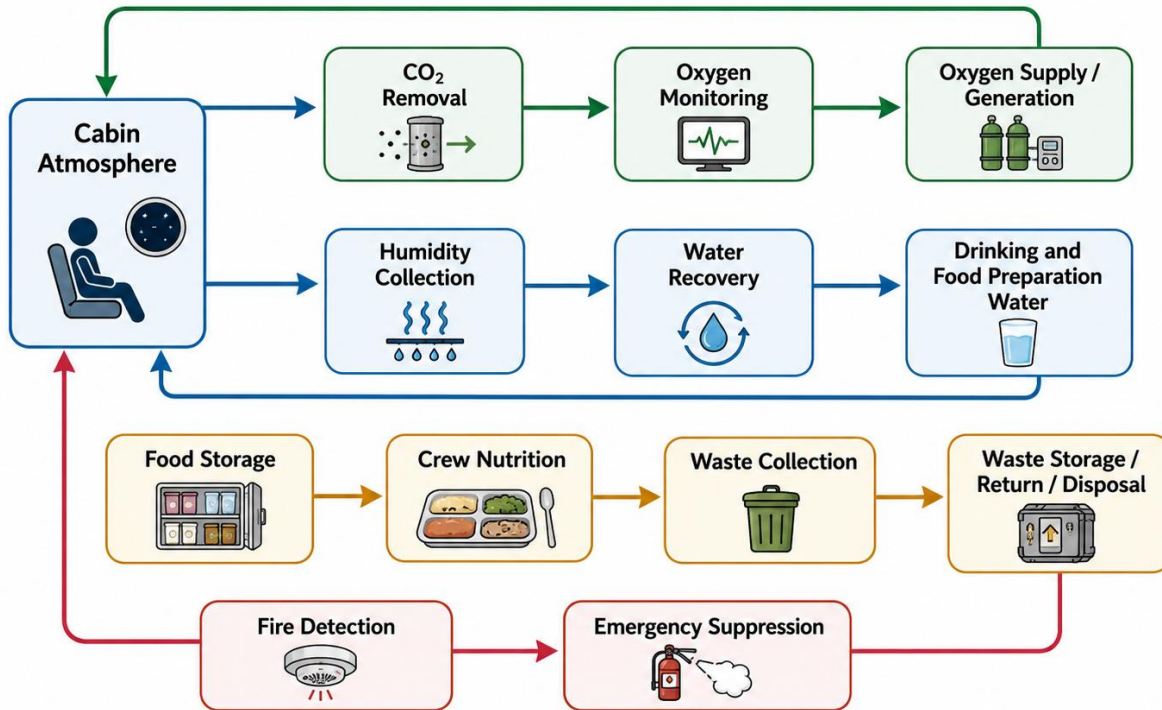
Internal layout concept for a compact solo spacecraft. The astronaut's seat is positioned near the radiation storm shelter and emergency storage to maximise survivability during solar particle events or cabin anomalies.

Life Support, Consumables and Redundancy

NASA's current public figures provide a practical starting point for consumables. ISS ECLSS documentation states that each crew member needs about **one gallon of water per day**, and NASA reports a per-person oxygen demand of roughly **1.83 lb/day**. For food, NASA mass-balance work gives a planning figure near **2.39 kg/day of individually packaged food delivered per person**, including packaging and menu margin. Over a **21-day one-person mission**, a purely open-loop baseline therefore implies roughly **79.5 kg water, 17.4 kg oxygen**, and about **50 kg packaged food**, before waste containment and medical reserve. [28]

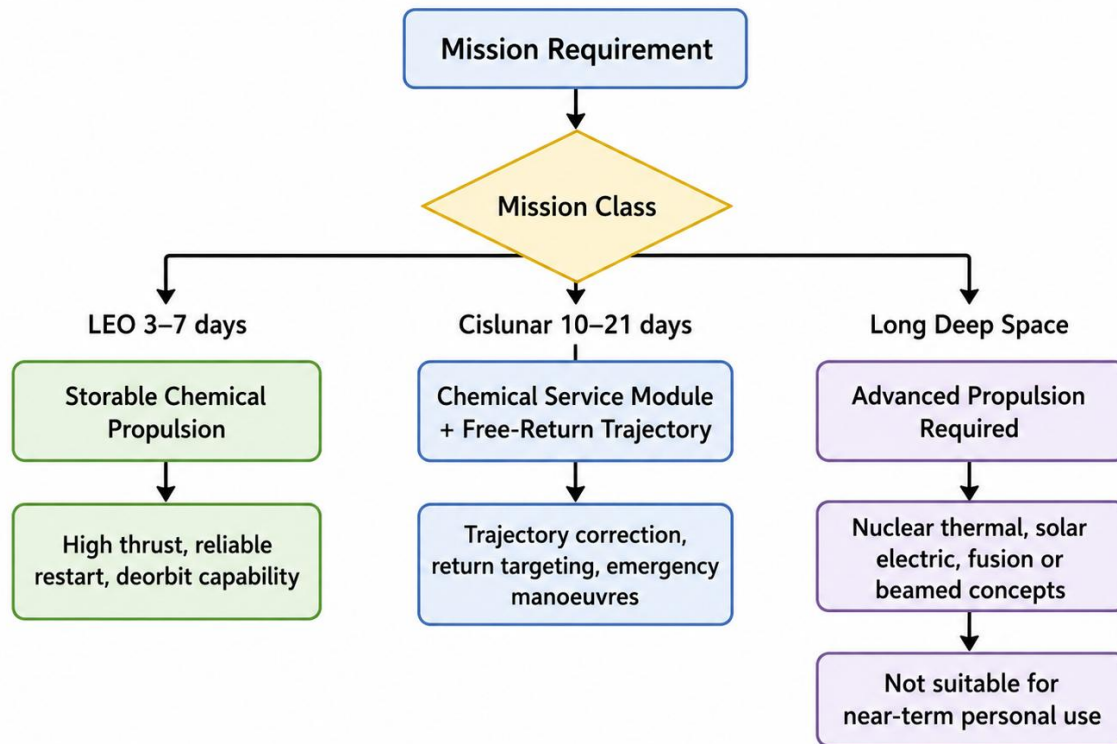
These figures make a decisive systems point. For **3–7 day LEO missions**, open-loop or lightly regenerative life support is likely lighter, cheaper, and more reliable than a fully regenerative loop. For **10–21 day cislunar missions**, *partial closure* becomes attractive, especially water recovery, humidity condensate recovery, and CO₂ scrubbing with limited oxygen regeneration. Full ISS-class closure remains technically possible, but Jones's life-cycle-cost analysis strongly suggests that the major development and hardware burden of deep recycling is justified on mission durations far longer than those analysed here. [29]

The solo-spacecraft ECLSS should therefore follow a three-layer approach. The **first layer** is open-loop survival consumables sized for the nominal mission plus reserve. The **second layer** is partial-loop recovery, especially water and humidity management. The **third layer** is fault tolerance: duplicate atmosphere sensors, independent fan paths, redundant CO₂ removal paths where feasible, fire detection and suppression, and a fully manual safe mode with checklists that do not assume constant ground support. NASA's human-system standards explicitly foreground vehicle design and operations requirements intended to maintain astronaut safety and performance; the solo case only strengthens the case for austere, verifiable redundancy. [30]



Simplified environmental control and life-support system for a solo spacecraft; for short missions, partial regeneration is more practical than a fully closed-loop system.

Figure 3. Life-Support System for One-Person Space Travel
Simplified environmental control and life-support system for a solo spacecraft.
For short missions, partial regeneration is more practical than a fully closed-loop system.



Chemical propulsion remains the most realistic near-term option because it provides immediate thrust, restart capability, and emergency manoeuvring.

Figure 4. Propulsion Decision Framework for a Personal Spacecraft
Propulsion trade logic for solo spacecraft missions. Chemical propulsion remains the most realistic near-term option because it provides immediate thrust, restart capability, and emergency manoeuvring.

Navigation, Guidance, Communications and Autonomy

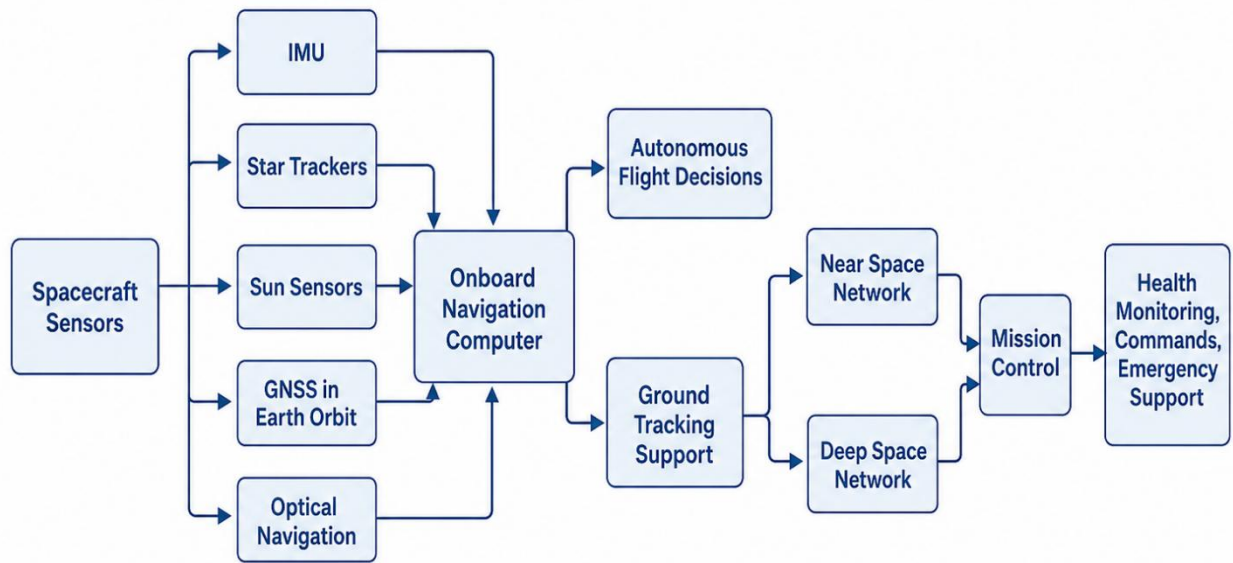
In Earth orbit, mature navigation practice fuses **IMU, Sun sensors, star trackers, and GNSS**. NASA’s 2026 GNC assessment describes this sensor fusion as standard for modern spacecraft. However, the same source makes clear that **deep-space navigation is performed using radio transponders in conjunction with the Deep Space Network**, and that future autonomy will need to reduce dependence on Earth-based tracking. For a solo spacecraft, this means that onboard autonomy is not a convenience but a safety requirement. [15]

GNSS is useful but not unlimited. NASA’s cislunar GNSS work states that the formal altitude limit of GNSS usage in space is **36,000 km**, while practical use can extend well beyond that with weak-signal reception from opposite-side satellites and sidelobes; nevertheless, geometry, Earth occultation, and signal weakness constrain performance. The correct design stance is therefore to treat GNSS as a powerful aid in LEO and high Earth orbit, a possible supplementary source in cislunar space, and **not** the sole

navigation backbone beyond GEO. A solo spacecraft aimed beyond LEO should carry star trackers, optical navigation capability, inertial sensing, and communications compatible with NSN/DSN or equivalent services. [31]

Relative navigation for docking or proximity operations should rely on camera-based and lidar-based sensing, because autonomous rendezvous and docking is already recognised by NASA as critical in both LEO and lunar-orbit mission concepts. IDSS-compatible docking adds not only physical compatibility but also the possibility of rescue, logistics transfer, or refuge. This is especially important for a solo craft, because the absence of an onboard second crewmember makes external assistance more rather than less important. [32]

Communications planning must also be conservative. NASA describes the DSN as the world’s largest telecommunications system for interplanetary spacecraft and notes that it, alongside the Near Space Network, supports lunar and deep-space missions. A solo spacecraft does **not** imply solo mission operations; even a single-occupant vehicle depends on tracking, command uplink, health monitoring, navigation support, weather and recovery planning, and anomaly response teams on Earth. “Alone” can mean a crew of one onboard, but not a mission operated by one person in a systems sense. [33]



Low-Earth-orbit missions can use GNSS heavily, while cislunar and deep-space missions require star tracking, optical navigation, onboard autonomy, and ground network support.

Figure 5. Navigation and Communication Architecture
Navigation and communication system for a solo spacecraft. Low-Earth-orbit missions can use GNSS heavily, while cislunar and deep-space missions require

star tracking, optical navigation, onboard autonomy, and ground network support.

Power, Thermal Control and Radiation Protection

Power architecture is one of the least controversial elements of the design trade. NASA's small-spacecraft power assessment notes that solar power generation is the predominant method for spacecraft, with rechargeable batteries as the norm, and that available solar energy follows the inverse-square law with distance from the Sun. For LEO and cislunar missions near 1 AU, a solar-array-plus-battery architecture is therefore the baseline choice. [34]

A reasonable notional electrical budget for the cislunar solo reference vehicle is shown below.

Table 3. Author-derived notional power budget for the one-person cislunar reference vehicle. It is intentionally scaled below Orion's 11 kW four-person capability but well above minimalist smallsat power levels, reflecting the continuous burden of life support and crew habitability. [35]

Electrical load	Average power	Peak power
ECLSS	650 W	900 W
Thermal control	450 W	900 W
Avionics and computing	180 W	250 W
GNC sensors and actuators	120 W	500 W
Communications	120 W	800 W
Lighting, habitability, hygiene, galley	150 W	250 W
Docking and proximity-ops support	80 W	400 W
Margin and contingency reserve	350 W	350 W
Total	2.10 kW	4.35 kW

On that basis, the spacecraft should target roughly **5–6 kW** generation at 1 AU and **12–20 kWh** usable battery storage, depending on eclipse duration and contingency margins. Such sizing provides headroom for communication peaks, docking, heaters, and partial autonomy during array off-pointing. Solar-only architectures become less favourable for much deeper interplanetary missions as flux falls with heliocentric distance, but they remain entirely credible for LEO and cislunar operations. [36]

Thermal control combines passive and active systems. NASA's thermal-control materials emphasise that spacecraft temperature is set by absorbed, stored, generated, and rejected

heat, and highlight ongoing development of multifunctional thermal structures, heat spreading, and passive methods. For a solo crew capsule, the baseline should be multilayer insulation, selective coatings, heat pipes or loop heat transport where required, compact radiators on the service module, and thermostatically controlled heaters for eclipse and survival modes. [37]

Radiation is a sharper discriminator between mission classes. NASA now uses a unified astronaut career radiation exposure limit of **600 mSv**, while also applying event-specific and mission-specific standards and ALARA principles. NASA human-performance material further notes that shielding is effective against **solar particle events**, but typically reduces **galactic cosmic ray** exposure by only about **7–15%**. Deep-space transit measurements associated with the Curiosity cruise measured an average of about **1.8 mSv/day**, a level that becomes serious over long interplanetary durations. For that reason, a cislunar solo craft should include a compact **storm shelter** built around water bags, food stores, and equipment mass near the seat or sleeping position, but a Mars-class solo mission remains radiologically unattractive with present methods. [38]

EVA, Docking and Rescue Interfaces

The docking interface should follow the **International Docking System Standard**, whose stated purpose includes enabling collaborative operations and possible crew rescue. NASA's docking materials also emphasise that IDSS-derived interfaces support automated docking and transfer of data and power. For a solo vehicle, this means docking is not merely a convenience for station visits; it is a primary risk-control feature. [39]

A pressure suit should be treated primarily as **launch, entry, and depressurisation survival equipment**, not as a full operational EVA system. NASA notes that modern suits are designed to sustain crew in the unlikely event the spacecraft loses pressure. This supports a prudent solo-spacecraft doctrine: carry integrated suit protection and contingency cabin-loss resilience, but do not build the mission around scheduled solo spacewalks. [40]

Mission Profiles and Operations

Feasible Mission Classes

The three mission classes considered here do not share equal feasibility. A **short LEO sortie** is the most credible near-term solo mission because communications are low-latency, rescue and deorbit are comparatively quick, GNSS is fully usable, and human exposure to radiation and microgravity is limited. A **cislunar free-return mission** is technically harder but strategically attractive because a free-return trajectory reduces propulsion demand and preserves a passive path back toward Earth. An **interplanetary**

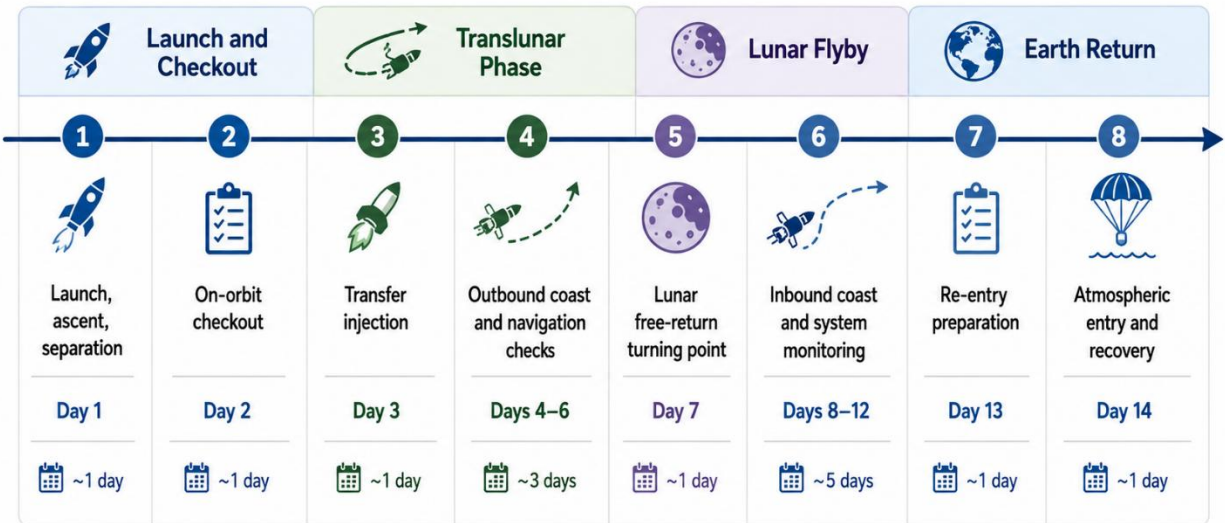
solo mission is presently not recommended because mission time, communication latency, and medical autonomy burdens shift the system from a capsule problem to a habitat problem. [41]

Table 4. Mission-class assessment for single-occupant spacecraft. Ratings synthesise NASA communications and autonomy work, DSN dependence, radiation evidence, and human-system constraints. [42]

Mission class	Nominal duration	Communications regime	Rescue and abort potential	Technical feasibility	Recommended status
LEO solo sortie	3–7 days	Continuous or near-continuous Earth support	Fast deorbit and recovery possible	High with current technology	Primary development target
Cislunar free-return	10–21 days	NSN/DSN-supported, modest delay	Limited; docking refuge strongly preferred	Moderate with current technology	Stretch target
Lunar orbit/landing solo	14–30+ days	More demanding than free-return	Complex; no simple passive return	Low-to-moderate	Only after cislunar free-return maturity
Mars-transfer solo	180–300+ days one way	Minutes to tens of minutes delay; blackouts	No practical rescue	Poor with present practice	Not recommended

Mission Timeline and Operational Logic

A cislunar free-return mission is the most interesting upper bound because it remains within the envelope of a compact spacecraft while avoiding the full complexity of lunar orbit insertion, surface operations, or long-duration deep-space habitation. It should still be understood as a professionally staged mission with launch, separation, checkout, trajectory corrections, outbound cruise, lunar swingby, inbound cruise, re-entry targeting, and recovery coordination. [43]



A free-return trajectory is preferred because it gives the spacecraft a passive path back toward Earth if propulsion capability is reduced.

Figure 6. Notional mission timeline for a one-person cislunar free-return profile. The sequence reflects standard crewed operations logic and deliberately prefers passive-return geometry over more aggressive mission plans. [44]

Even for LEO, “independence” must be qualified. Flight operations require launch-site and range integration, licensing compliance, tracking, mission control, medical support, weather and splashdown or landing planning, public safety analysis, and mishap-response procedures. FAA launch and re-entry licensing practice, NASA’s ISS operations culture, and DSN/NSN communications doctrine all reinforce the same lesson: **solo occupancy does not eliminate system-of-systems dependence.** [45]

Interplanetary Operations and Why They Fail the Current Solo Test

The interplanetary case fails not because propulsion is unimaginable, but because the integrated human-system burden grows beyond what a small solo craft can responsibly contain. NASA states that Mars communications delays are on the order of **4–24 minutes one way**, with possible blackouts around solar conjunction; NASA autonomy work similarly stresses that crews beyond LEO will confront reduced evacuation options, reduced resupply, and delayed expert support. Meanwhile, deep-space radiation remains chronic and only partly shieldable, and exercise demands increase markedly with mission duration. These factors turn a small spacecraft into an under-sized habitat with insufficient resilience. [46]

Safety, Human Factors, Medical Contingencies and Risk

Human Factors and Habitability

NASA's human-spaceflight standards treat human factors, habitability, and environmental health as primary system requirements, not afterthoughts. For a solo spacecraft, this means the cabin must support cognitive performance, sleep, hygiene, restraint, visibility, manual override, acoustic control, and tolerable workload in both nominal and off-nominal conditions. The risk of single-person error is higher because the usual crew cross-check is absent; interface design must therefore bias toward simple display logic, aggressive fault isolation, and clear procedural branching. [47]

Food, lighting, schedule discipline, and exercise are not comfort accessories but safety elements. NASA's public human-systems guidance specifies an average energy requirement around **12,698 kJ/day (3,035 kcal/day)** per crewmember unless mission-specific values are derived, and mass-balance work for missions beyond **30 days** recommends about **90 minutes of exercise per person per day**. Even though a 10–21 day solo mission may be shorter than classic long-duration thresholds, the habitability design should still provide a minimal exercise capability, a reliable meal-heating and rehydration method, and circadian-friendly lighting and workload planning. [48]

Psychological factors are also intensified in a crew-of-one mission. NASA's hazard literature on distance from Earth emphasises the operational burden created by communication delay and reduced immediate support. A one-person spacecraft therefore needs more than a seat and controls; it needs structured routines, alertness management, pre-scripted decision support, and anomaly procedures designed for degraded cognition under stress. This is another reason why mere geometric miniaturisation of Mercury or an ultralight capsule is an inadequate model for modern solo flight. [49]

Medical Contingencies and Autonomous Care

Medical contingencies are among the strongest arguments against long solo missions. NASA's autonomous medical care literature notes that exploration missions face limitations in equipment, supplies, and expert support, while medical emergencies may be difficult or impossible to evacuate. NASA's evidence review on adverse outcomes and autonomous care further highlights that long-duration missions impose mass and volume constraints, limited crew medical expertise, and delayed communications. For a lone occupant, those constraints are more severe because there is no medically capable crewmate. [50]

A credible solo-spacecraft medical approach therefore requires at least four components: rigorous pre-flight screening and training; an onboard emergency medical kit sized to the mission class; diagnostic support tools such as guided ultrasound or equivalent compact diagnostics where practical; and software-based medical decision support with

asynchronous telemedicine when signal latency allows. Even with those measures, the acceptable mission duration remains limited, because *self-treatment under hypoxia, trauma, cardiac symptoms, or neurovestibular compromise is intrinsically unstable*. This is one of the clearest reasons that interplanetary solo flight is presently indefensible. [51]

Risk Matrix

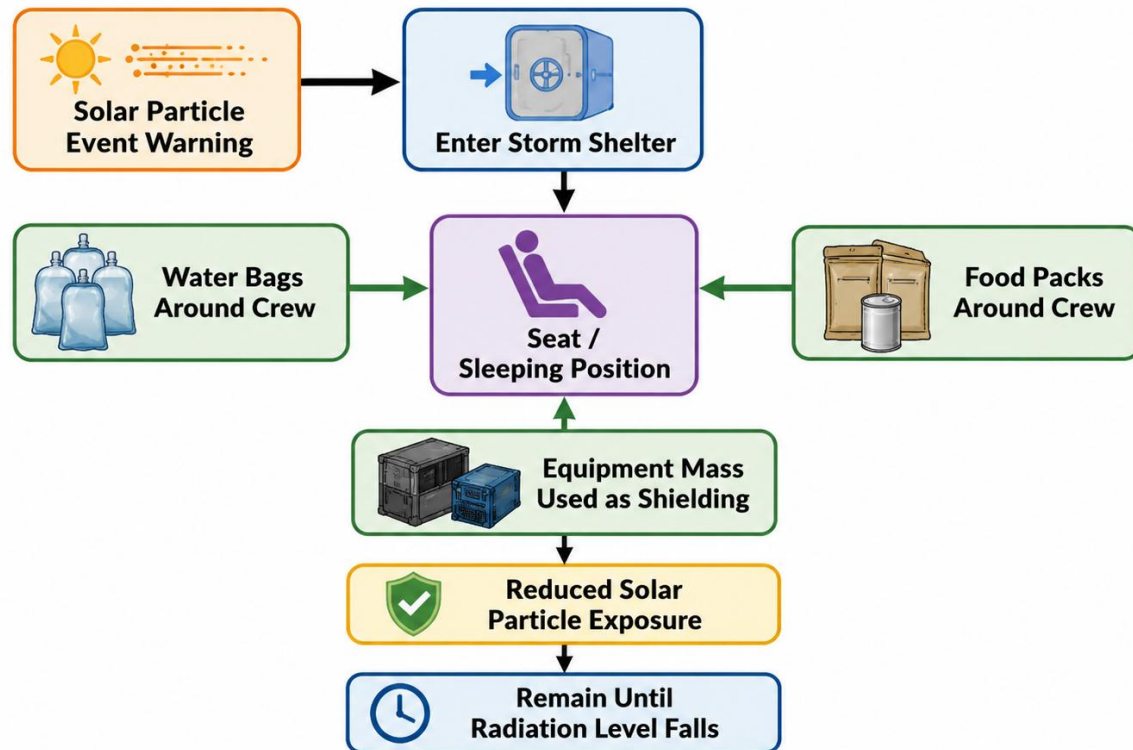
The following matrix is an **authorial synthesis** based on the cited technical, medical, and regulatory literature. It expresses qualitative mission risk rather than published probabilistic risk assessment.

Table 5. Qualitative risk matrix for solo human spaceflight. Hazard classes reflect the cited literature on human-system design, autonomy, radiation, EVA, relative navigation, and legal responsibility. [52]

Hazard	Likelihood	Consequence	Overall risk	Principal mitigation
Launch or ascent failure	Low–moderate	Catastrophic	Very high	Certified launcher, abort system, crew suit, escape and recovery planning
Cabin depressurisation	Low	Catastrophic	Very high	Pressure integrity, suit backup, leak isolation, no nominal EVA
Fire or toxic-atmosphere event	Low	Catastrophic	Very high	Fire-safe materials, rapid detection, suppression, isolated ventilation paths
Power-system failure	Low–moderate	Major to catastrophic	High	Solar-array redundancy, batteries, load shedding, thermal-safe mode
GNC or software loss of control	Moderate	Major to catastrophic	High	Independent sensors, FDIR, manual safe mode, ground abort authority

Hazard	Likelihood	Consequence	Overall risk	Principal mitigation
Docking/proximity-ops collision	Low–moderate	Major	High	Lidar/camera relative nav, autonomous keep-out zones, no rushed docking
Radiation SPE during cislunar mission	Low	Major to catastrophic	High	Storm shelter, dosimetry, solar monitoring, mission timing
MMOD strike	Low	Major to catastrophic	High	Whipple shielding, attitude management, short mission duration
Medical emergency	Moderate	Major to catastrophic	High	Screening, autonomous medical kit, telemedicine, short mission profile
Communication loss beyond LEO	Moderate	Major	Medium–high	Onboard autonomy, stored procedures, independent navigation, robust comms

The dominant pattern is that many risks remain **low in likelihood but catastrophic in consequence**, which is the classic signature of crewed spaceflight. That pattern strongly undermines the idea that a single person could casually or privately “travel freely” in space without institutional infrastructure. A safer interpretation is that a professional solo mission might be allowed in a narrow envelope where rescue time, comms delay, and consumables remain bounded. [53]



Existing onboard mass such as water, food, and equipment can be arranged around the astronaut to reduce exposure during solar particle events.

Figure 7. Radiation Storm Shelter Concept

Radiation storm shelter concept using existing onboard mass such as water, food, and equipment as shielding around the astronaut during solar particle events.

Law, Regulation, Insurance, Cost and Ethics

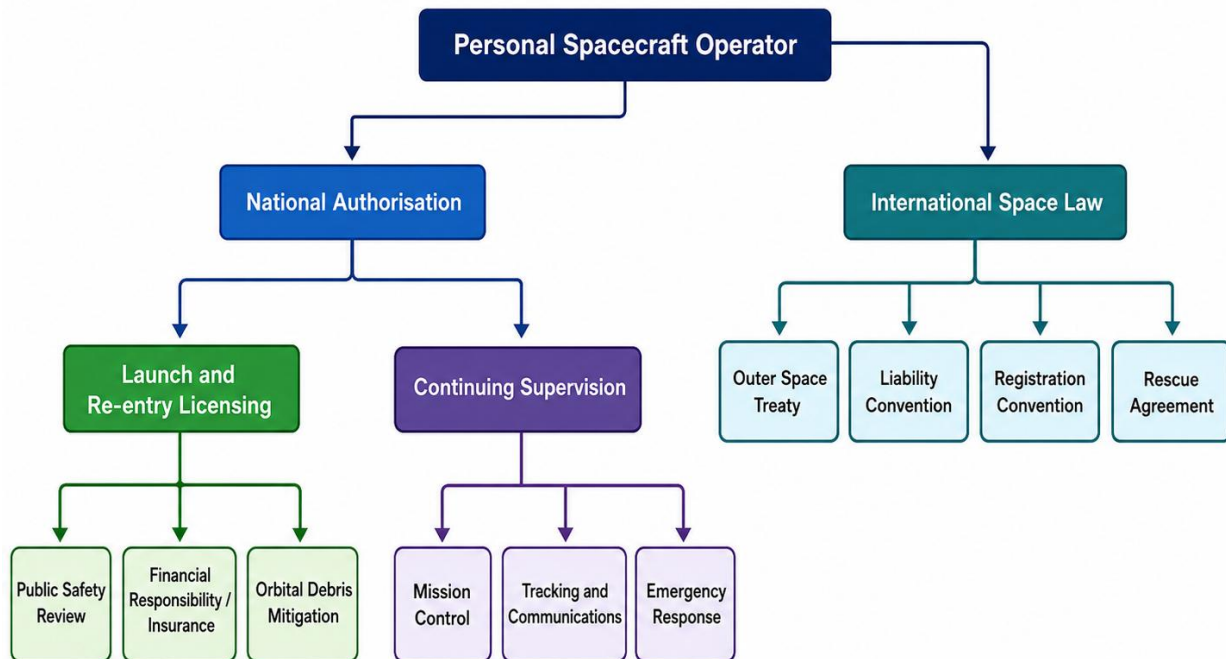
Legal and Regulatory Framework

The legal architecture is unequivocal. Outer space is free for exploration and use by states on a basis of equality, but **private actors are not exempt from state control**. Article VI of the Outer Space Treaty requires that non-governmental activities be authorised and continuously supervised by the appropriate state; Article VII establishes launching-state liability; Article VIII links jurisdiction and control to the state of registry; and Article IX imposes duties of due regard, consultation, and avoidance of harmful contamination or interference. The Rescue Agreement, Liability Convention, and Registration Convention reinforce those obligations. A solo spacecraft therefore cannot exist in a legal vacuum. [54]

As a practical regulatory example, the FAA requires licences or permits for commercial launches and re-entries involving US operators or operations in the United States, with Part 450 covering launch and re-entry operator licensing and Part 440 governing financial responsibility. FAA human-spaceflight rules also address crew and spaceflight participant matters, and the statutory “informed consent” framework remains part of commercial human spaceflight practice. Even where another national licensing regime applies, the FAA example demonstrates the depth of public-safety, licensing, and oversight expectations likely to burden any would-be personal spacecraft programme. [55]

Orbital debris and end-of-life disposal are equally material. The US Government Orbital Debris Mitigation Standard Practices aim to limit debris released during normal operations, minimise on-orbit break-ups, and reduce post-mission derelicts; NASA passivation guidance emphasises removing stored energy at end of mission. ESA’s contemporary debris work similarly stresses disposal, collision avoidance, and preventing the persistent occupation of orbits. Thus, any “personal” vehicle in LEO would require not only a launch licence, but also a credible disposal, deorbit, and passivation plan. [56]

For interplanetary missions, planetary protection also enters the framework. COSPAR’s planetary-protection policy remains the principal international reference standard, and Article IX of the Outer Space Treaty obliges avoidance of harmful contamination. A truly free solo interplanetary mission with uncertain end-of-mission disposal would therefore be ethically and legally problematic even before the crew-safety analysis is considered. [57]



Personal spacecraft operations require both national supervision and compliance with international space law because launch, re-entry, debris, liability, and rescue obligations cannot be ignored.

Figure 8. Legal and Operational Control Framework
Governance model for solo personal spaceflight. A personal spacecraft cannot operate outside national and international supervision because space activity involves launch safety, re-entry risk, orbital debris, liability, and rescue obligations.

Insurance and Financial Responsibility

Insurance for a solo spacecraft is unlikely to resemble conventional personal-vehicle insurance. In the US framework, licensees must demonstrate financial responsibility, typically via insurance or other financial means, based on maximum probable loss calculations. Public FAA material underscores that financial responsibility is an integral part of launch and re-entry licensing, and that payload review also considers hazard, disposal, and international obligations. The immediate implication is that insurability is partly a consequence of licensing, and for a one-off solo human spacecraft it may be difficult to separate hull risk, third-party liability, and state-backed indemnification structures. [58]

This means the insurance question is not merely actuarial but structural. A vehicle that carries a human, manoeuvres independently, and re-enters through national airspace

imposes liabilities on the operator, insurer, launch provider, and state of registry. For that reason, a “personal spaceship” is more likely to be insured, if at all, as a specialised commercial human-spaceflight activity rather than as a privately owned recreational asset. [59]

Cost Estimates and Budget Envelope

The cost case is the most decisive non-technical barrier. NASA reports that Commercial Crew Transportation Capability awards totalled **US\$6.8 billion** across Boeing and SpaceX, while GAO reported in 2025 that Orion’s estimated development cost was **nearly US\$10 billion**. These figures are not directly transferrable to a one-person vehicle, but they reveal the correct order of magnitude for certification, integration, and human-rating in modern orbital crew transport. A single-occupant craft would be smaller, but the certification, abort, operations, and oversight burden does not scale linearly with seat count. [60]

Public launch pricing provides only a lower bound. SpaceX’s current public Falcon 9 pricing lists **US\$74 million** per launch, and its rideshare pricing lists **US\$350,000 for 50 kg to SSO** with additional mass at **US\$7,000/kg**. Those numbers are useful because they anchor present-day launch-market scale, but they do **not** represent the full cost of a solo human mission, which would additionally require a licensed human-rated mission profile, mission control, recovery, training, hazard analyses, and likely bespoke integration. [61]

A defensible order-of-magnitude budget therefore looks as follows.

Table 6. Order-of-magnitude budget ranges for solo-spacecraft development and operation. These are analytical estimates anchored to official public programme costs and launch pricing, not market quotations. [62]

Cost element	Solo LEO demonstrator	Solo cislunar demonstrator	Comment
Spacecraft development and qualification	US\$1.5–4.0 bn	US\$3–10 bn	Anchored by Commercial Crew and Orion public programme costs
Ground systems, mission software, control and recovery	US\$0.1–0.3 bn	US\$0.3–0.7 bn	Includes operations infrastructure and simulations
Human-rating, abort and safety certification effort	US\$0.2–0.8 bn	US\$0.5–1.5 bn	Strongly coupled to national licensing and test campaigns

Cost element	Solo LEO demonstrator	Solo cislunar demonstrator	Comment
Recurring spacecraft build/refurbishment per flight	US\$30–120 m	US\$80–250 m	Highly design-dependent
Launch services per flight	> US\$74 m lower bound	Likely much higher or multi-element	Cargo-price analogue only, not a human mission quote
Mission operations, training and reserves per flight	US\$15–50 m	US\$30–100 m	Excludes exceptional insurance events

The plain conclusion is that a real solo spacecraft is **not personal in a consumer-market sense**. Even an aggressive derivative design would require institutional capital, deep systems engineering capability, and regulatory engagement far beyond normal private means. Economically, the most plausible path would be not personal ownership of a wholly novel vehicle, but purchase of access to a professionally operated, heavily standardised single-occupant mission service. [60]

Ethical Considerations

Ethically, three principles dominate. The first is **non-maleficence to the occupant**: there is a difference between consenting to risk and exposing a single human to a mission architecture that cannot reasonably provide rescue, medical support, or radiation mitigation. The second is **non-maleficence to others**: debris generation, bad re-entry planning, and poor traffic coordination impose risk on other operators and the public. The third is **fair use of a shared domain**: the Outer Space Treaty explicitly frames outer space as the province of all humankind, and Article IX requires due regard to the interests of others. [63]

These principles have concrete design implications. The spacecraft should not be permitted to loiter as debris. It should not rely on EVA when robotics or docking can avoid it. It should not pursue interplanetary trajectories without planetary-protection compliance. And it should not interpret “freedom” as a licence to externalise operational risk onto rescue services, insurers, or other spacecraft operators. [64]

Conclusions, Future Work, Declarations and References

Conclusions

The answer to the research question is both affirmative and restrictive. **Yes**, a lone human can technically travel in space in a dedicated spacecraft, and the historical and contemporary literature is sufficient to sketch a rigorous, one-person reference design. **No**, that capability is not “free” in the ordinary sense, and it is not presently reducible to the ownership model of a private aircraft or boat. The mission remains bounded by propulsion, consumables, life-support architecture, radiation, communications, rescue doctrine, law, and cost. [65]

The most publishable engineering conclusion is that the **correct near-term target** is a **minimal but professionally operated LEO solo capsule**, able to support 3–7 days independent flight, launched on an already certifiable infrastructure, and designed with no nominal EVA. The **correct second-step target** is a **10–21 day cislunar free-return craft** with docking compatibility, storm-shelter mass allocation, partial-loop life support, and substantial autonomy. A truly solo **interplanetary** spacecraft should, on present evidence, be regarded as outside the responsible design envelope. [66]

The most important future work is not a new propulsion advertisement but a multidisciplinary closure campaign. Needed advances include better compact radiation shelters and materials, stronger Earth-independent autonomy, compact medical diagnostics and treatment support, validated one-person human-factors models, and programme-level standards for small-crew or single-crew rescue doctrine. Without those advances, “personal space travel” remains best described as a narrow expeditionary capability rather than a personal mobility revolution. [67]

Declarations

Authorship.

The author declares that this research paper is an independent conceptual and technical study prepared for educational and academic discussion.

Conflicts of interest.

The author declares no known conflict of interest. Just out of curiosity of the space travel.

Funding.

This research received no external funding.

Data availability.

No new experimental dataset is introduced. All factual inputs are derived from the cited public or official sources; all budgets and architecture figures marked as notional are analytic estimates produced for this manuscript.

Ethical Statement.

This paper does not involve human subjects, animal testing, clinical data, or private personal data.

Disclaimer.

This document is a research-style feasibility study and should not be interpreted as a safety certification, flight readiness review, or legal opinion. [68]

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